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Load Balancing of Smart Agriculture Data Using Edge Computing and SDN

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Introduction

- The use of IoT revolutionized agricultural practices, enhancing efficiency, productivity, and sustainability. The application of IoT (Internet of Things) in smart farming includes (M. Ayaz, M. Ammad-Uddin et. Al, 2019) Autonomous irrigation management, Crop selection, Livestock Monitoring, Crop management, Climate control, Nutrient level management, Predictive Analytics and Decision Support, Energy Management.

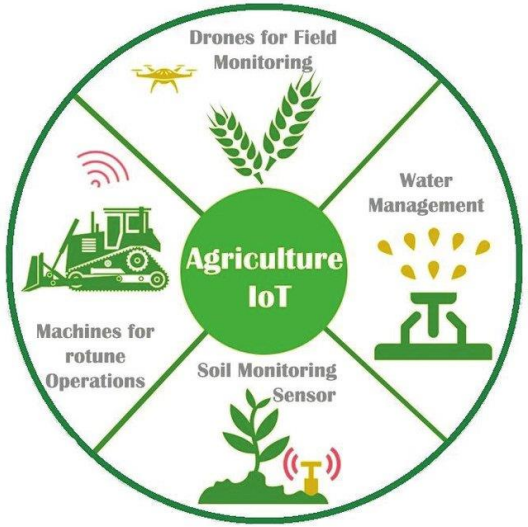


Fig. Smart farming practices (Swamidason, I. T. J. ., Pandiyarajan, 2022)

Traditional IoT networks still faces big challenges during data transmission at cloud server in various smart agricultural application (Quy, V.K.; Hau et al., 2022).

- Cloud computing does not have the potential to support all the data that will be needed to process it in real time for decision making and manage all various network.

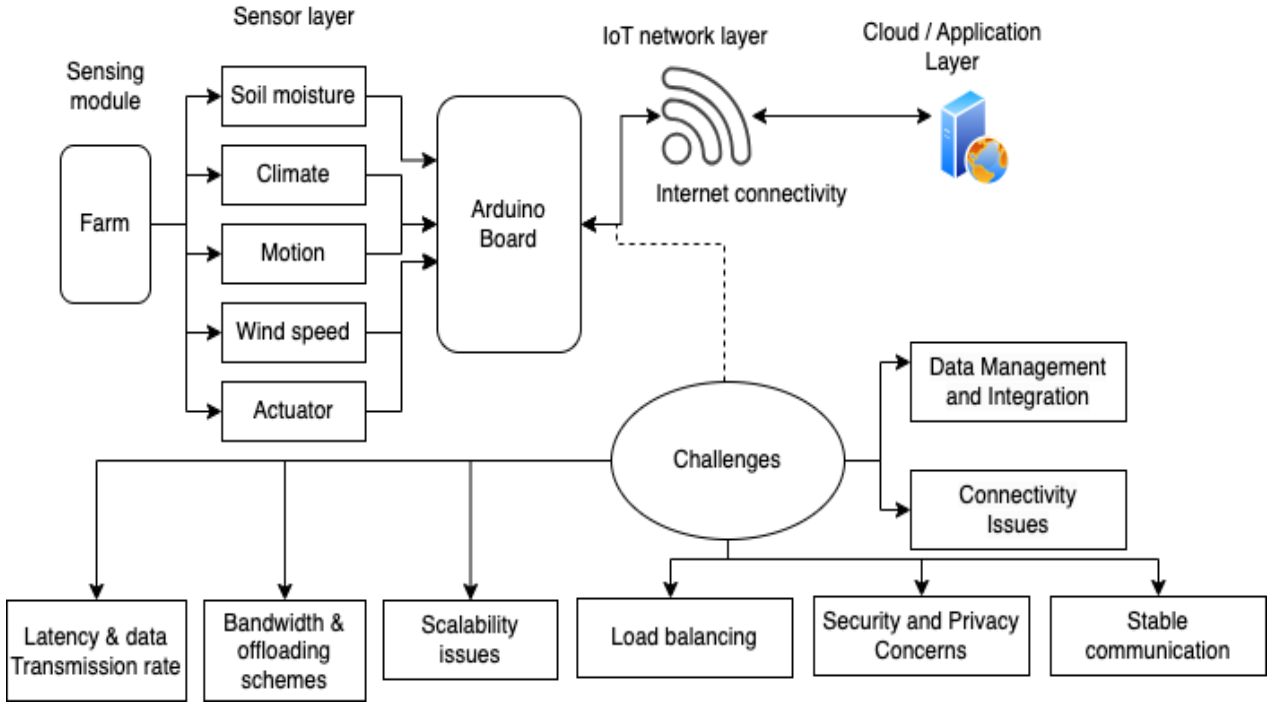


Fig. IoT enable farming system with issues

Introduction (cont..)

- Edge computing has the scalability to support this type of problem to become a form of information management.
- Edge computing able to connect everything with everyone. Edge computing can also be traced to 2002 when concepts such as distributed networks were used, where a system was proposed with more than 12,000 servers in more than 1000 networks (J.Dilley,etal, 2002, Ruiz-Vanoye, Jorge et al., 2022).

Objective

- Store, organize, and process the data extracted from agro IoT devices at the edge network layer and transmit the useful data with equal priority at the cloud server.
- Enabling the user to obtain real-time information more efficiently by connecting a decentralized edge-based server as an edge layer with the nearest edge devices using LoRaWAN at each farming field. (R. S. Alonso et al,. 2020)

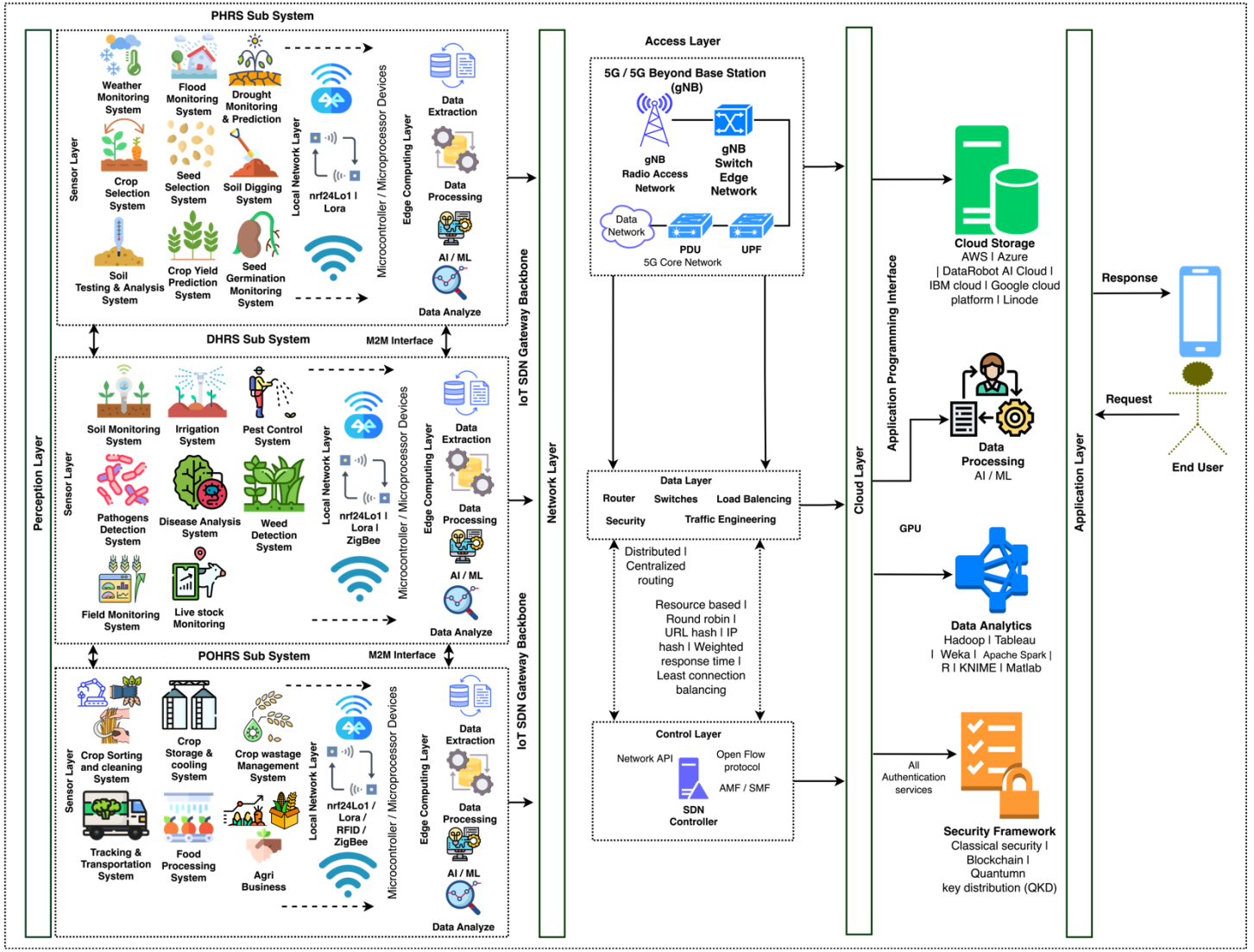


Fig. Distributed IoT Edge computing smart agriculture framework (Kasera, Rohit Kumar, et al., 2024)

Exciting problem & Novel contribution (cont..)

- Cutting-edge computing collaboration
- Sufficiently adaptable to handle large farming fields Edge computing data processing.
- Distributed edge network nature
- Seamless integration
- Flexibility of data transmission
- SDN responsiveness according to network conditions
- Compatibility
- Reliability and Connectivity



Fig: Latency

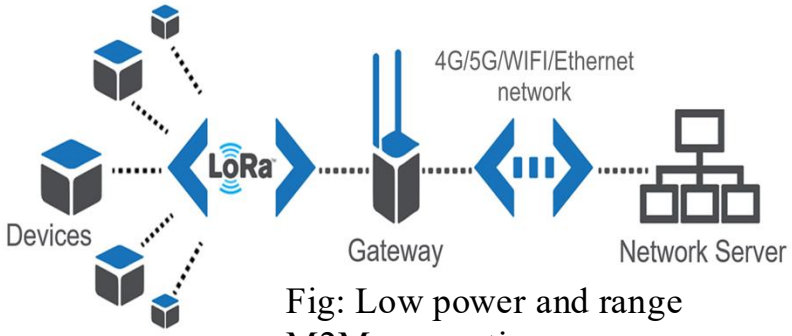


Fig: Low power and range M2M connection

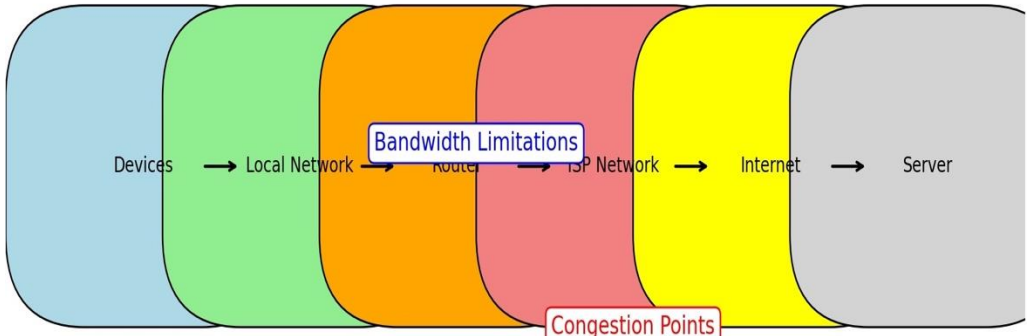


Fig: Network congestion

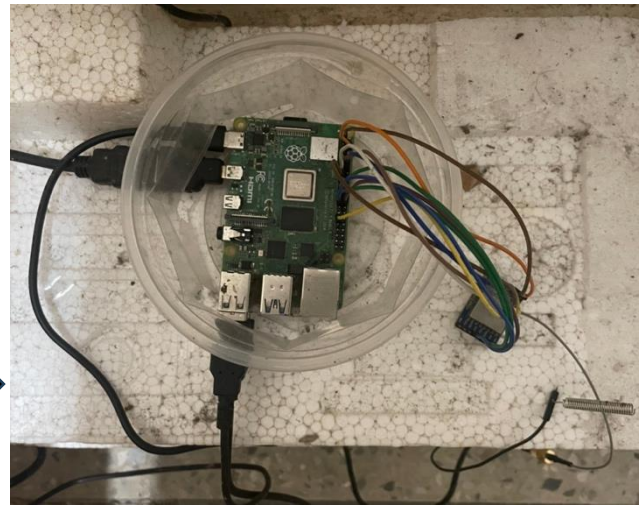
- Real-time streaming of data across distributed edge devices and edge gateways is facilitated by a long-range (LoRa) 433 MHz module over multiple or large farming fields.
- Developed a low-power wide-area network (LPWAN) Edge SDN-based framework for data pre-processing and cleaning, load balancing of the distributed Agro edge-based IoT application using a modified round-robin mechanism, as Equitable and Efficient Resource Distribution (EERD) that oversees the precise and informed data transmission between edge servers and a cloud.
- Raspberry pi as distributed edge server and Mininet simulator is used to create virtual topology network.
- SDN server and edge network is controlled using Ryu controller.

Methodology

- LoRa-based communication playing a vital role for the IoT based system to make wide area network, low energy usage, bidirectional in nature, protected, reliable networks, massively distributed nodes in the network, and maximum long-range transmission coverage (S. Devalal, et. al, 2018). This requirement is met by using the wireless LoRa 433 MHz module to transmit data from different farming fields to their nearest edge server.
- Multiple edge gateway load are managed through the proposed Equitable and Efficient Resource Distribution (EERD) an SDN-based load-balancing algorithm which is based on the round robin approach. Load balancing rules and real-time edge device monitoring, allocation of resources with fairness are made simpler by this centralization.
- Raspberry pi is used as edge server which is used to receive the field sensor data through LoRa sx1278 433 Mhz module.
- SDN server is created Ryu controller and miniNet simulator is used to configure virtual topology node.



Fig: Open farming field LoRaWAN enabled sensing device (Transmitter)



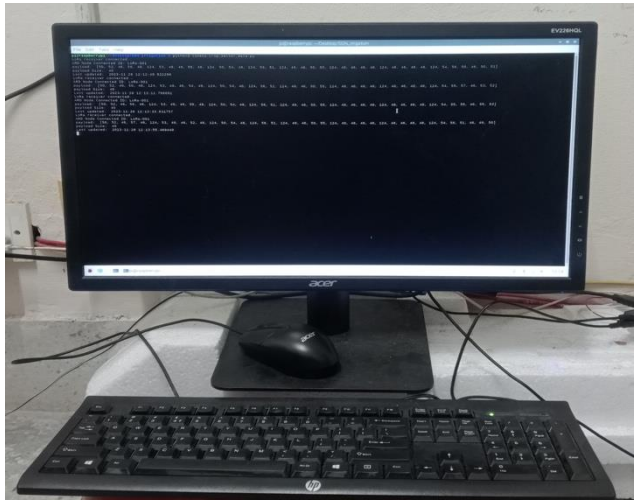


Fig: LPWAN Edge Gateway with SDN enabled (Receiver)

Algorithm

- $Edge_n \rightarrow PI_1, PI_2, ..., PI_n$ is the aggregate amount of LPWAN edge server devices as raspberry pi.
- $A_i \rightarrow$ allocation of resources for PI_i in which $1 <= i <= Edge_n$
- DS_i is the data size for the Pi_i
- RRQ → The order of devices in the queue.
- $EDSM_i \rightarrow$ Monitoring PI_i Ds_i
- MaxAR → Maximum accessible resources
- $\Delta RC_i \rightarrow$ More resources allocated for optimizing DS_i
- The controller method is used to allocate resources with fairness to connect the user load with the gateway and manage the nodes using OpenFlow-based switches that watch over traffic at the data plane.

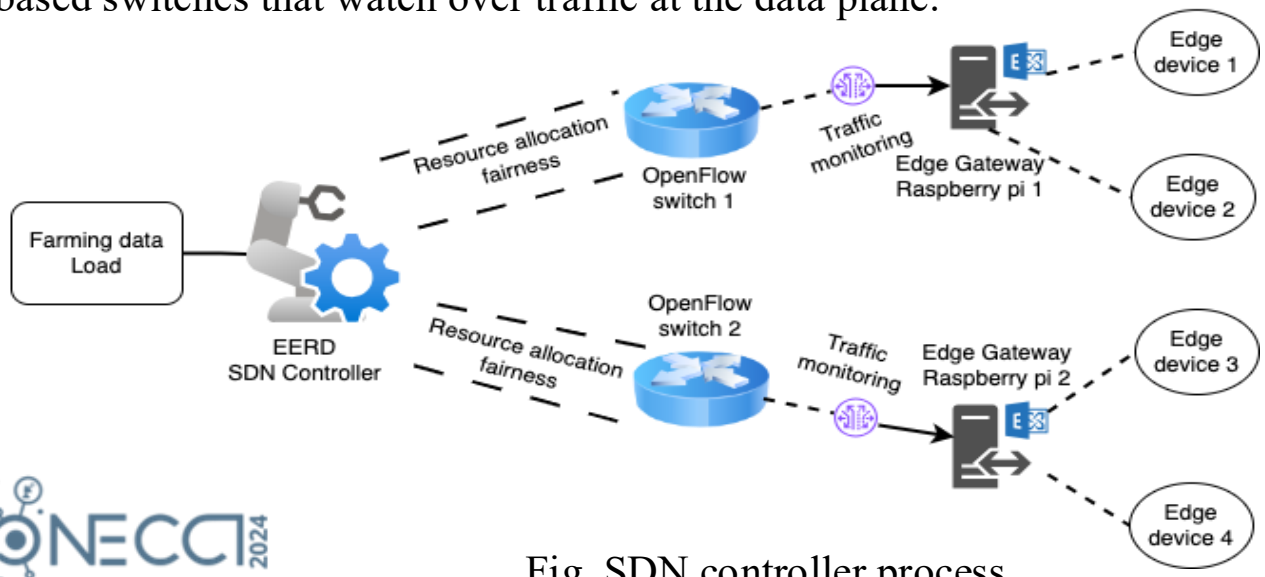


Fig. SDN controller process

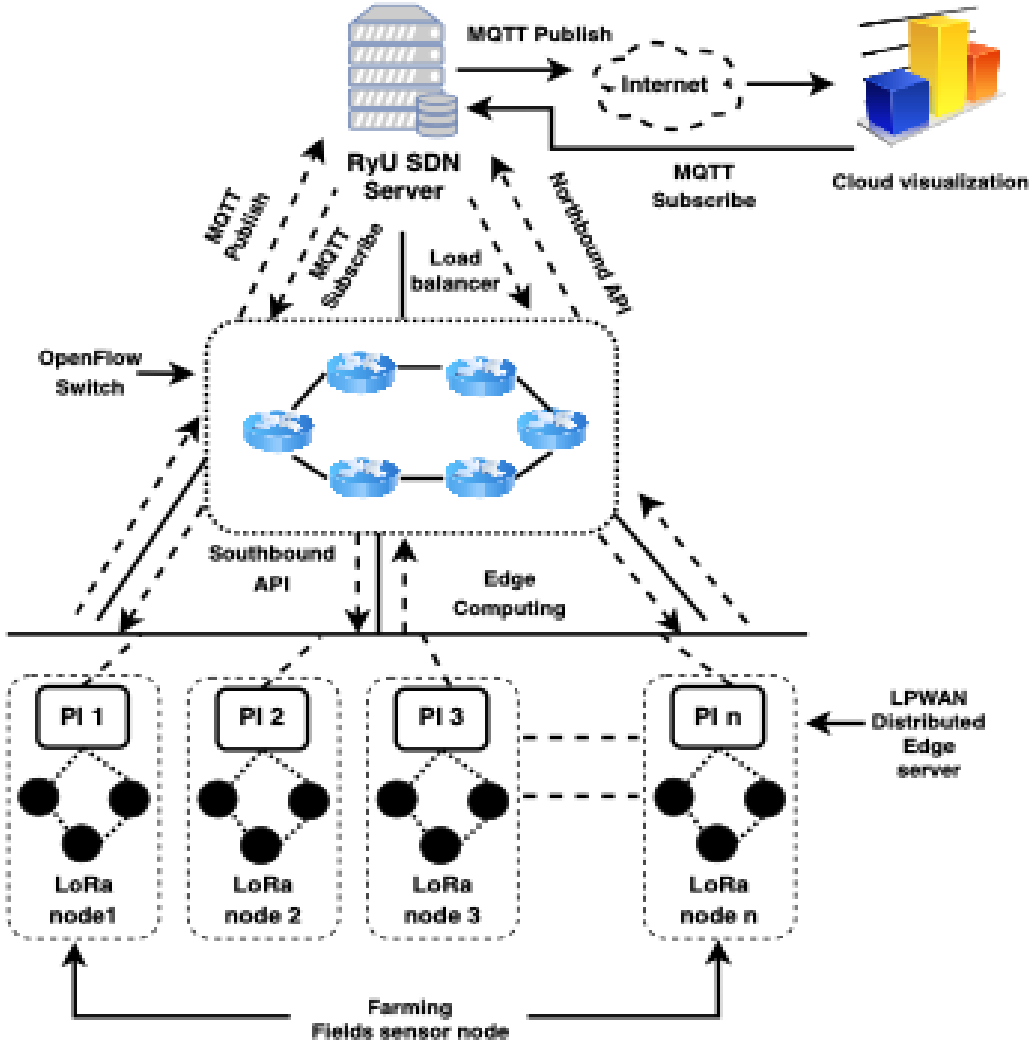


Fig. Distributed Edge SDN based smart farming system controller architecture

Algorithm (cont...)

- The LoRaWAN edge-based server uses the "Message Queuing Telemetry Transport" (MQTT) protocol to publish data to the cloud and continuously measure SDN environmental parameters like DS_i edge server information, and level of noise assessment.
- During the MQTT communication, SDN controller is used to monitor this network traffic and send feedback to each edge server on the successful transmission of data with allocated resources, data lost or received, transmission time, etc.

Data preprocessing (Dp_i) = $f_i(ud)$ (1)

- f_i → edge server's evaluating function, is used to pre-processing, cleaning, and computing tasks.
- ud → unstructured data collected from edge device.

Objective method

$$EERD_{Max} = \sum_{i=1}^{Edge_n} Usefulness\ (Dp_i, A_i, DS_i, EDSM_i, RRQ) \quad (2)$$

- Usefulness → acquire data size to determine resource allocation equitably and effectively as well as dynamic resource management depending on RRQ order.

$$A_i = \frac{MaxAR}{Edge_n} \quad (3)$$

A_i → Dynamic resource allocation and optimization.

$$\text{Remaining resources} = MaxAR - \sum_{i=1}^{Edge_n} A_i \quad (4)$$

$$\Delta RC = \min(Ds_i,\ MaxAR - A_i) \quad (5)$$

$$OAR = AR_i + \Delta RC_i \quad (6)$$

- To optimize DS_i data loss is prevented by allocating additional resources to the device by extent of its data

Algorithm……

Algorithm 1 Equitable and efficient resource distribution to allocate resources to an LPWAN edge gateway with equal priority

1. Input: $Edge_n$ [PI_1, PI_2,……., PI_n], MaxAR
2. Output: Monitoring of data size and allocated resources
3. RRQ → queue(1, to $Edge_n$ +1)
4. DSM[]→ for each PI_i in RRQ
5. **while true** do
6. for PI_i in RRQ do
7. PI_id → RRQ.remove.PI_i.id /*remove temporarily*/
8. Measure remaining resources using equations (4)
9. allocate_resources(PI_id, remaining resource, MaxAR)
10. optimize_resource_distribution(PI_id, EDSM[PI_id], ΔRC)
11. RRQ.add (PI_id)
12. **end for**
13. allocated_resource_tracking()
14. Mqtt_get_msg()
15. **end while**
16. Function allocate_resources (id, remaining resource,MaxAR)
17. A → using equation (2) and (3)
18. if (AR > residualResource) then
19. OD → residualResource
20. else
21. OD → AR
22. end if
23. Dv[id] → OD
24. MaxAR → MaxAR - OD
25. return OD
26. end
27. Function optimize_resource_distribution (PI,EDS,residualResource)
28. if (residualResource <= 0) then
29. None resources available to allocate
30. end if
31. Calculate additional resources ΔRC using equations (5).
32. Connect SDN to allocate resources to a PI device
33. Remaining resource → residualResource – ΔRC /*Update remaining resources after allocation*/
34. end

Results and Discussion

- The experiment, data was continuously transmitted from each edge gateway to the cloud over MQTT. The SDN controller was utilized to monitor and manage the network traffic during this transmission.
- The "LoRa" transceiver was deployed to communicate with the edge gateway, which provide an aggregate links rate of 280 kilobytes per second (kbps) at short range and 30.29 kbps at long range more than 12000 meters with spreading factor of 7 to 12.
- 93.87% is the average rate at which resources are allocated to each edge gateway.
- The average response time of the edge server in EERD-based load balancing is found to be 91.23 MS overall, but in the round-robin and without load balancing method, the average response time is measured to be 112 MS and 126 MS, which analyze a slower response time and delay time was 1.35 MS.

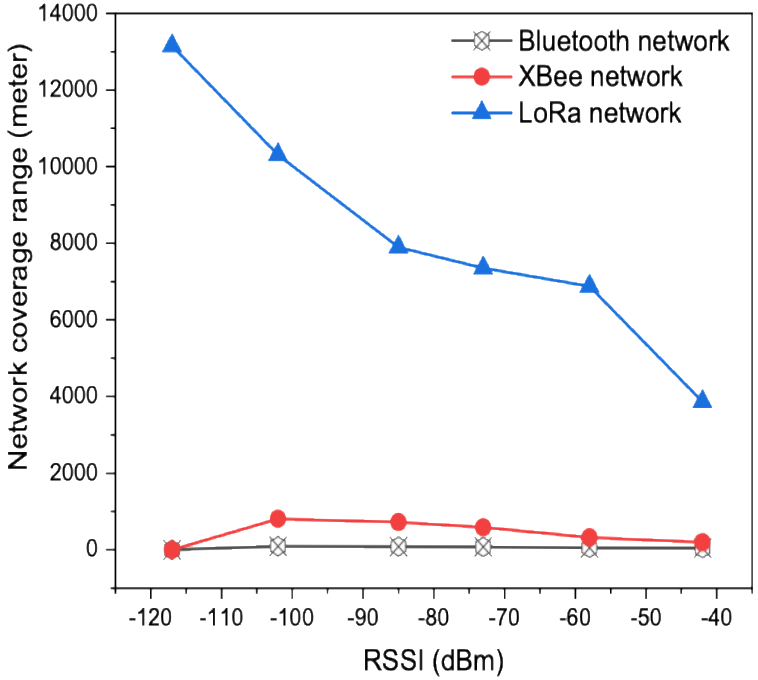


Fig. Comparative LPWAN analysis with another network

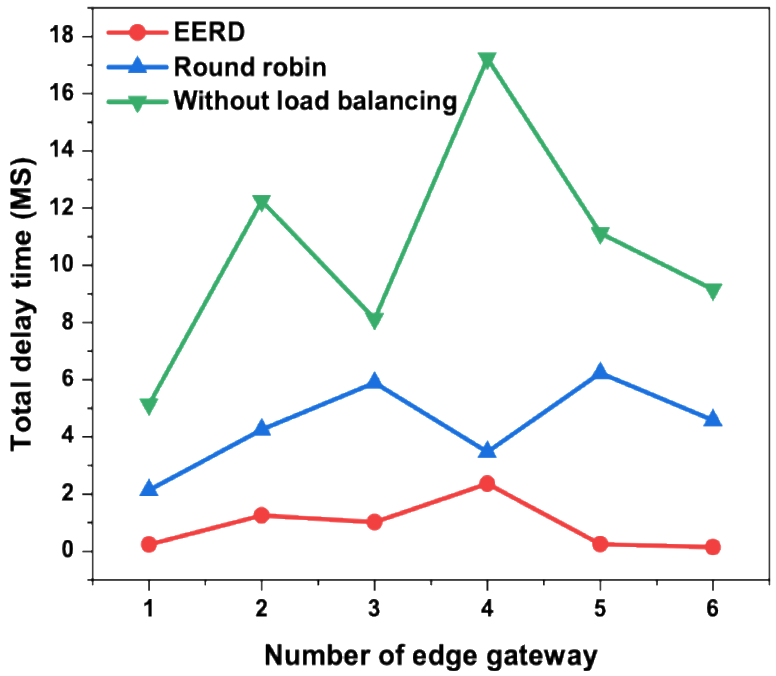


Fig. Delay time analysis

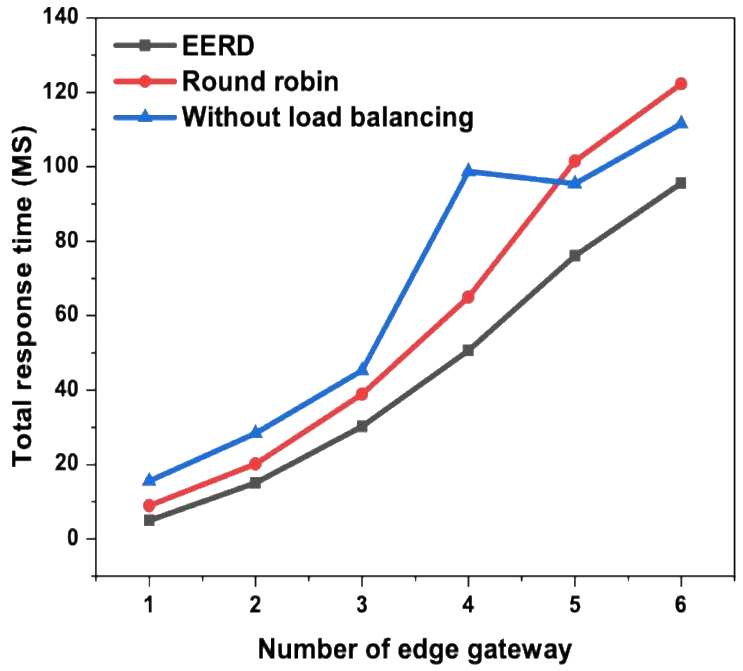
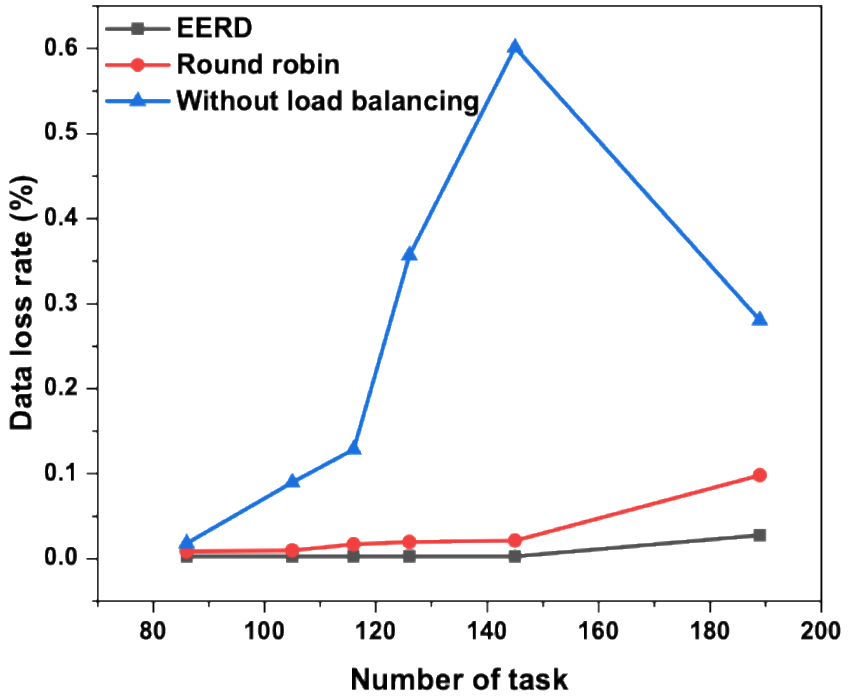
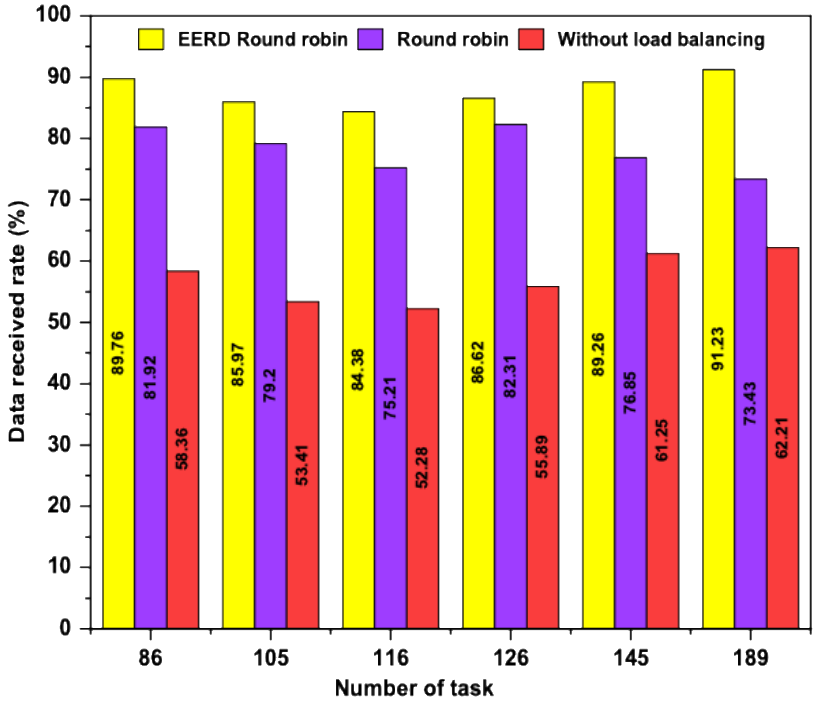


Fig. Response time analysis

Results and Discussion (cont…)

The estimated total average data loss and the received rate are 0.076% and 88.12%





- IoT-based agricultural sensor data being processed at the LPWAN SDN edge node increases storage capabilities locally and prevents congestion while data is transmitted to the remote cloud.
- MQTT messaging allows data to be routed between data planes between SDN edge gateways and the cloud while minimizing delay time, reducing data integrity risks, and managing real-time access.
- The proposed EERD edge SDN method could leverage the use of LoRaWAN technology and SDN to effectively load balance data traffic, specifically in environments where IoT edge devices are distributed across multiple farming areas.

- The data to the cloud while dynamically with equitability handing out resources and prioritizing numerous edge gateway data streams.

Performance Parameter	EERD	RR	Without load balance
Average delay time (ms)	2.01	5.05	9.5
CPU consumption	9.2	6.7	6.4
Throughput (kbps)	88.03	87.23	88.98

Conclusion

- To reduce overload, latency duration, and data degradation and to enable instantaneous access to agriculture field-related data, the approach, as EERD had been proposed to fairly distribute the resources with equal priority to a distributed edge server.
- The overall experiment results and discussion that the proposed EERD-based algorithm performs better in allocating resources dynamically in comparison to the traditional round-robin approach based on the CPU consumption, the data size, the delay time, the throughput, and the number of events.

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Thank You